

these intermediate results to a solution for the original problem it is trying to solve.

The standard textbook example for MapReduce is to count the number of occurrences of each word in the input. The map phase starts with parsing a section of input and emitting a set of key-value pairs such as <word, 1> for each occurrence of the word it sees. The key-value pairs can then be sorted and passed to a set of reducers, which combine/reduce the same word occurrences into the total count of occurrences of each word. The key thing to note is that both the Map and Reduce phase are parallel operations—a number of mappers can run in parallel, and so can a number of reducers. MapReduce also has the ability to restart computation in case any of the worker nodes fail, as well as supporting data reliability by replication. Hadoop MapReduce is nothing but the open source implementation of the MapReduce paradigm, using a special-purpose distributed file system known as HDFS. Hadoop is an open source project from Apache (www.apache.org) that provides a software framework for reliable, scalable distributed computing.

Hadoop is meant to be scalable, designed to run on infrastructure ranging from a large supercomputer to a cluster of standard PCs, each offering local computing capabilities and local storage. Instead of depending on custom hardware to achieve reliability in large-scale distributed computations, reliability is built into the software library, with the assumption that hardware failures are likely to occur, and it is the software's responsibility to handle such failures. The Hadoop open source project consists of three basic components: Hadoop Utilities, Hadoop MapReduce Framework and HDFS. Hadoop also has a number of sub-projects built on it, such as HBase, a scalable distributed database for structured data; Hive, a data warehouse built atop Hadoop; and ZooKeeper, a co-ordinated service for distributed computations atop Hadoop. This article will cover just Hadoop's MapReduce framework. Let us look at the various offerings based on Hadoop's MR framework and its alternatives.

Hadoop offerings in the market

There are a wide variety of Hadoop offerings in the market. There are the pure Hadoop distributions such as Cloudera (www.cloudera.com/), HortonWorks (www.hortonworks.com/), EMC's Hadoop version called GreenPlum HD, and MapR's Hadoop distribution (www.mapr.com/). These are packaged software distributions for setting up and running Hadoop on clusters based on the open source ASF Hadoop project. These packaged distributions of Hadoop facilitate its easy adoption, instead of the user having to download the different packages needed to set it up from www.apache.org/hadoop. These Hadoop distributions are often integrated

with other Data Warehousing solutions or MPP databases, so that data movement can occur seamlessly between Hadoop clusters and the other data management solutions. For instance, as part of their AWS, Amazon offers the Elastic MapReduce Service, which delivers an enterprise-grade Hadoop platform on the cloud on top of a standard ASF Hadoop distribution.

A number of enterprise vendors give Hadoop offerings on top of their data management solutions. For example, EMC's GreenPlum Unified Analytics Platform combines GreenPlum's Massively Parallel Database meant to handle the structured data; a Hadoop distribution based on ASF, known as Greenplum HD; and Chorus, a productivity and group-ware layer for data science teams. By allowing data movement and interoperability between the Greenplum database and Hadoop, it is possible for a query to access both sources of data, and combine the information retrieved from these two sources in an intelligent manner.

IBM offers its own Hadoop solution, which is known as InfoSphere *BigInsights* (www-01.ibm.com/software/data/infosphere/biginsights/). This is under IBM's InfoSphere big data information management brand of offerings. IBM's *BigInsights* supports the Jaql query language. This supports interoperability of the Hadoop offering with other IBM products such as their enterprise data-warehousing solution IBM Netezza, and their RDBMS DB2. *BigInsights* also includes various Hadoop enhancements, including cluster administration and management tools, as well as textual analysis tools.

Microsoft also offers a Hadoop distribution integrated with its Windows Server product, as well as with its cloud platform Azure (www.microsoft.com/sqlserver/en/us/solutions-technologies/business-intelligence/big-data-solution.aspx). Microsoft has developed a Javascript API for its Hadoop distribution, which allows the developer to write MapReduce tasks in Javascript. The company also provides the integration of Hadoop with Visual Studio and .NET to enable developers to rapidly develop Hadoop-based applications on the Windows platform.

Oracle, too, has jumped on the Hadoop bandwagon by offering a Hadoop appliance (www.oracle.com/us/products/database/big-data-appliance/overview/index.html). This appliance includes Hadoop, R for analytics, a new Oracle NoSQL database, and connectors to Oracle's database and Exadata data warehousing product line. Its Hadoop distribution is from Cloudera. Many of the existing MPP (massively parallel processing) databases that are specialised for processing structured big data also offer Hadoop connectors. For instance, Aster Data MPP (which was bought over recently by Teradata), ParAccel MPP and Vertica MPP (which was acquired by HP) offer Hadoop connectors. There are also a number of Hadoop management software solutions, to manage and troubleshoot Hadoop implementations. Many of the Hadoop distributions